

HEALTHCONNECT CLINIC

EXPERIENCE LAB

Week 4 Project Summary — Data Analytics Track

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

1. Specific Problem Addressed

The Data Analytics contribution focuses on understanding appointment attendance and no-show patterns within the HealthConnect appointment dataset. The initial objective is to assess the structure and quality of the available data, identify variables that may be relevant to appointment outcomes, define meaningful business questions and KPIs, and establish a structured approach for deeper analysis in subsequent weeks.

2. Resources Used

The following project resources were reviewed:

- The Week 4 HealthConnect Experience Lab Assignment Brief.
- The HealthConnect Appointment Dataset.
- The HealthConnect Data Dictionary.
- The HealthConnect Clinic Knowledge Base.

3. Key Observations from Week 4

The HealthConnect appointment dataset contains 5,000 appointment records and 18 variables.

The `booking_date` and `appointment_date` variables were identified for conversion to datetime format to support chronological analysis.

Three variables were identified as containing missing values. `reminder_channel` contained 1,366 missing values, representing 27.32% of the dataset. A cross-tabulation with `reminder_sent` showed that all 1,366 missing reminder channel values corresponded to appointments where no reminder was sent.

This indicates that the missingness in `reminder_channel` is structural rather than necessarily resulting from incomplete data. Where no reminder was sent, a reminder delivery channel does not apply.

The `distance_to_clinic_km` variable contained 90 missing values, representing 1.80% of the dataset. The `waiting_time_minutes` variable contained 60 missing values, representing 1.20%. These variables have relatively limited missingness and require further assessment before decisions are made regarding imputation or exclusion.

4. Relevant Business Questions

- How does appointment type relate to appointment outcomes?
- Does receiving an appointment reminder relate to appointment attendance?
- Which reminder channels are associated with the highest appointment attendance rates?
- How do previous no-shows relate to the outcome of a current appointment?

- How does distance from the clinic relate to appointment outcomes?

5. Potential KPIs

Appointment Attendance Rate: Measures the proportion of appointments resulting in attendance.

Appointment No-Show Rate: Measures the proportion of appointments missed and directly addresses the central attendance problem.

Reminder Effectiveness Rate: Supports comparison of attendance outcomes between appointments where reminders were sent and where no reminders were sent.

Attendance Rate by Reminder Channel: Supports evaluation of the association between reminder channels and appointment attendance.

Attendance Rate by Previous No-Show History: Supports investigation of whether previous attendance behavior is associated with current appointment outcomes.

6. Proposed Analysis Approach

Data preparation and validation: Validate data types, investigate missing values, check duplicate records and appointment identifiers, and validate categorical values. Significant transformations will be documented, and cleaned data will be maintained separately from the original dataset.

Descriptive analysis: Examine the distribution of appointment outcomes and key appointment, reminder, historical and accessibility variables.

Relationship analysis: Investigate associations between appointment outcomes and appointment type, reminder status, reminder channel, previous no-shows, distance to clinic and waiting time.

Visual exploration: Use appropriate charts for categorical and numerical variables, including outcome comparisons and a numerical correlation matrix. Correlation findings will be interpreted as associations and not evidence of causation.

7. Key Considerations and Limitations

- The dataset is fictional and anonymized for the Experience Lab, so findings should be interpreted within the context of the project dataset.
- The available variables may not capture every factor influencing appointment attendance.
- The remaining missing values in `distance_to_clinic_km` and `waiting_time_minutes` require further investigation before treatment decisions are made.
- Correlation and observational comparisons can identify associations but cannot independently establish causal relationships.

8. Proposed Week 5 Focus

Week 5 will focus on deeper exploratory analysis of appointment attendance and no-show patterns. The planned analysis will investigate relationships between appointment outcomes and appointment characteristics, reminder practices, previous attendance behavior and accessibility factors.

The aim will be to identify meaningful patterns that can support evidence-based recommendations for improving appointment attendance and reducing missed appointments.

Conclusion

Week 4 established the analytical foundation for the HealthConnect project. The dataset structure and key variables were reviewed, initial data quality issues were assessed, and the reminder_channel missingness was investigated and identified as structural in relation to records where no reminder was sent. Relevant business questions, potential KPIs and an initial analysis approach were also defined to guide subsequent stages of the project.